

PROBLEM SET 5 : WRAP-UP SESSION

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This wrap-up tutorial is meant to review some of the key ideas of the term (but this is not an exhaustive review). The idea is to make sure you know what you have to solve for. We don't provide full solutions, as they can be directly obtained from the slides or the book.

Country A

Consider a two-period open economy of a country called Country A, in which households have preferences given by

$$\log C_1 + \log C_2$$

where C_1 and C_2 denote the consumption of good X in periods 1 and 2. The country does not produce X , but can buy it in the international markets. Households are endowed in period 1 and 2 with Q_1 and Q_2 units of a good Y that is sold in the international markets. In both periods, X sells for one unit of Y . The economy starts period 1 with assets B_0 . The world interest rate is r^* and there is free capital mobility.

The government of the country is interested in exploring other open economies to learn from their experience. Let's start by studying the local economy.

1. Write down the budget constraint of the households in each period, and then, derive the household's intertemporal budget constraint.
2. Write down the optimization problem of the household.
3. Recently, the government received the good news of a significant improvement in students' educational performance, as measured by test scores. This improvement should positively impact the future productivity of workers, which in this economy would imply an increase in the endowment in period 2 Q_2 . Given that the economy is open, explain how will consumption react in period 1? Compare to what would happen in a closed-economy case.
4. The government is also interested in the effect a reduction in the world interest rate. What will be the sign of the income effect? For this part, assume that $B_0 < 0$.

Country B

Now consider a similar economy. In country B the households have the same preferences for consumption, but their government has incorporated a tariff in each period τ_1 and τ_2 . Using the revenue from the tariff, the government gives lump-sum transfers to the households (L_1 and L_2).

1. Derive the household budget constraint, the government budget constraint, and the economy's intertemporal resource constraint in equilibrium. Compare with the maximization problem in Country A.
2. Observing these results, Country A government asks you what is the effect of the tariff on households' consumption. What would you answer?

Country C

Now, consider the case of another economy, country C. Households in country C also value consumption in 1 and 2 as in country A, but as they suffer constantly from earthquakes, so that they do not know with certainty their endowment in period 2. Assume households start period 1 with $B_0 = 0$ and that they have only access to a single internationally traded bond that pays interest rate $r^* = 0$ in all situations (financial markets are incomplete).

Suppose an earthquake occurs with probability p . Q_1 is the endowment in period 1, Q_2^b is the endowment in period 2 if there is an earthquake and Q_2^g if there is no earthquake.

1. How will the households expected utility look like if they have similar preferences than in country A? Write down their period budget constraints.
2. Write down the optimization problem. What happens if $Q_2 = 0$? Explain the intuition of that case.
3. The government of country A is examining the effects of income volatility on consumption. They are particularly interested in identifying which economy will experience the greatest volatility in consumption: country C or country C with complete financial markets. What would you answer?

Countries D and E

Now, consider the case of countries D and E. In those countries, households only consume two goods, X and Y . Of those goods, only X is internationally traded. In country D, X is α times more expensive than Y . In country E, X is β times more expensive than Y .

1. Provide examples of what goods X and Y could be.
2. Explain how we could calculate the real exchange rate between those two economies. Can we with the information provided? How could we know which country is the most expensive?